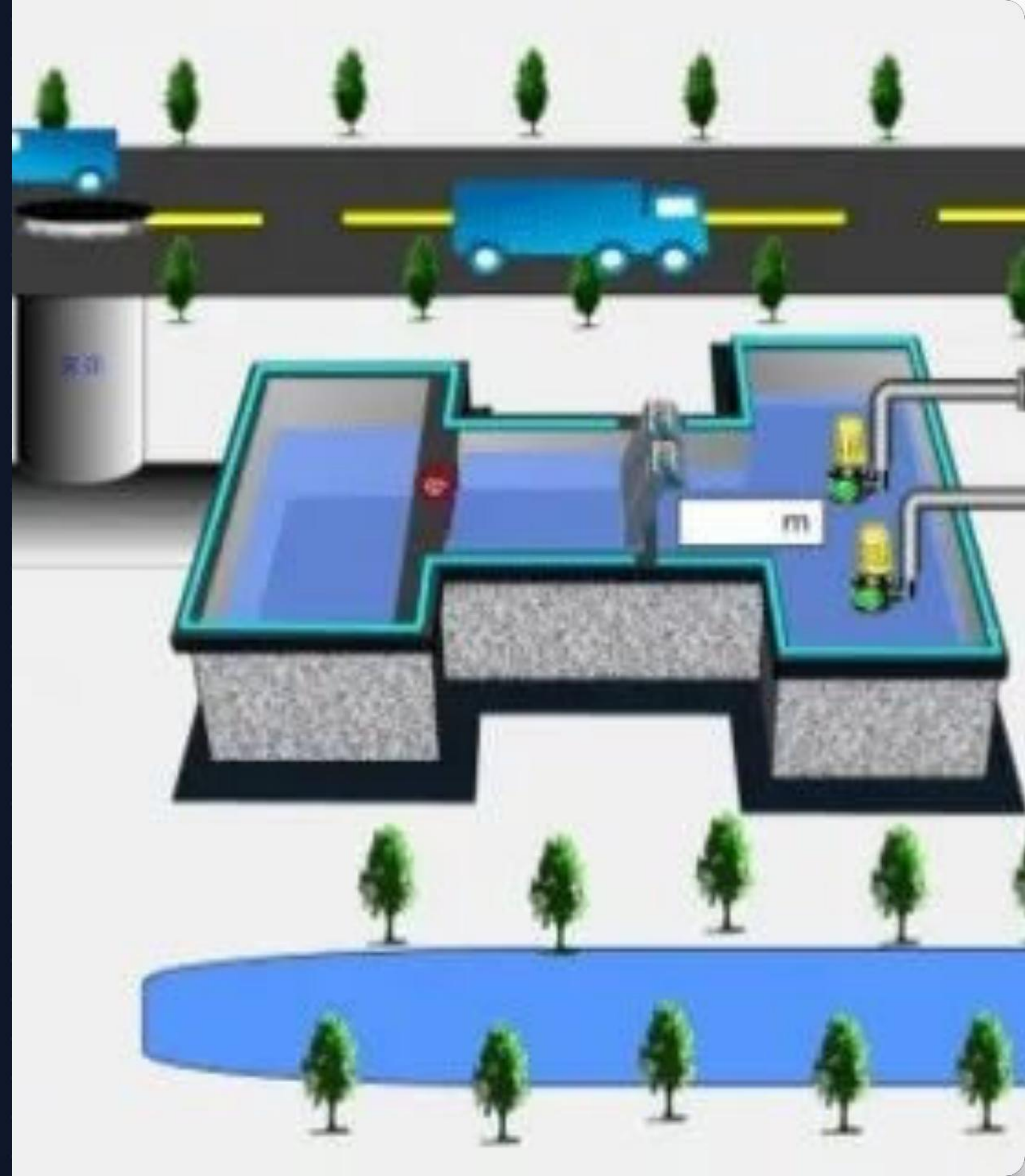


Smart Urban Drain Overflow & Flash Flood Warning Node

An IoT-enabled real-time drainage monitoring system designed for early flood warnings and municipal overflow detection.

Team : Error 404



Project Objectives & Benefits



Early Surge Detection

Monitors rapid water level rise rates ($\frac{\Delta h}{\Delta t}$) in roadside drains to trigger instant alerts before flash floods overflow onto streets.



Civic Flood Mapping

Provides a real-time, color-coded neighborhood map for commuters and local authorities to visualize drainage capacity during heavy rains.



Versatile Power Options

Supports flexible power input options—including standard 5V DC adapters, external battery packs, or mini solar PV panels for off-grid drain locations.

Project Tasks & Module Breakdown

Project Module	Core Activity & Scope	Key Output / Deliverable
Sensing & Hardware	Non-contact water level monitoring, signal filtering, and protective outdoor casing setup.	Weatherproof drain sensor node
Power Management	Power-saving sleep routines and integration of flexible power sources (adapter, battery, or solar).	Adaptable power subsystem
IoT Connectivity	Establishing cloud data transmission pipelines and automatic network reconnect handling.	Reliable telemetry link
Server & Alert Engine	Processing incoming depth data to calculate rapid surge rates and trigger emergency alerts.	Automated flood trigger engine
User Interface & Map	Developing an interactive live public map showing color-coded drain capacities and warning channels.	Web GIS map & alert bot

System Architecture & Flow

1. Non-Contact Sensing

Ultrasonic sensor measures drain water height continuously from under the top grate.

2. Edge & Power Control

Onboard processor filters surface waves, optimizes power cycles, and supports multiple power inputs.

3. Cloud Telemetry

Data streams to cloud servers to evaluate surge rates and threshold limits.

4. Community Alerts

Triggers instant Telegram broadcast notifications and updates the live public web map.



Measuring Project Benefits & Performance

> 15 Mins

Early Warning Lead Time

Evaluates how far in advance the system detects rapid rate-of-rise ($\frac{\Delta h}{\Delta t}$) to warn commuters before drains overflow.

± 5 mm

Distance Measurement Accuracy

Measures ultrasonic sensor precision under ambient variations and verifies digital noise filtering against water splash ripples.

< 5 Secs

End-to-End Alert Latency

Measures total time from water surge event detection on the node to emergency notification dispatch and web map updates.

Micro-Amps

Power Efficiency & Versatility

Evaluates deep-sleep power consumption and verifies seamless multi-source input compatibility (5V DC, battery packs, or solar PV).

Image Sources



<https://www.bivocom.com/wp-content/uploads/2023/03/pipe-network.jpg>

Source: www.bivocom.com



https://media.ncd.io/sites/2/20240703144056/Tank-Level-Sensor_A.jpg

Source: store.ncd.io